

Tobia Marcucci

Assistant Professor at the University of California, Santa Barbara and Amazon Scholar

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Research interests

My research lies at the intersection of convex and combinatorial optimization, with applications to robotics, motion planning, and control. I focus on optimal decision making in circumstances where discrete and continuous choices have to be made jointly. I work on these problems on a mathematical and computational level: I design efficient formulations as optimization problems and fast solution algorithms.

Employment

- **University of California, Santa Barbara** 3/2025 to present
Assistant Professor of Electrical and Computer Engineering
Affiliate of the Department of Mechanical Engineering
Affiliate of the Center for Control, Dynamical Systems, and Computation
- **Amazon** 7/2026 to present
Amazon Scholar

Education

- **Massachusetts Institute of Technology** 6/2018 to 5/2024
Doctoral Degree in Computer Science with minor in Mathematics
Advisors: Russ Tedrake and Pablo Parrilo
Thesis: [Graphs of Convex Sets with Applications to Optimal Control and Motion Planning](#)
Recipient of the [MIT EECS George M. Sprowls Ph.D. Thesis Award in AI and Decision Making](#)
- **University of Pisa** 12/2013 to 9/2015
Master's Degree in Mechanical Engineering
Final evaluation: 110/110 cum laude
- **University of Pisa** 11/2010 to 11/2013
Bachelor's Degree in Mechanical Engineering
Final evaluation: 110/110

Additional Research Experience

- **Amazon Robotics** 6/2024 to 2/2025
Postdoctoral Scientist
- **Stanford University** 11/2022 to 10/2023
Visiting Ph.D. student with Stephen Boyd
- **Massachusetts Institute of Technology** 1/2017 to 11/2017
Visiting Ph.D. student with Russ Tedrake
- **University of Pisa and Istituto Italiano di Tecnologia** 9/2015 to 1/2018
Ph.D. student with Antonio Bicchi (uncompleted, moved to MIT)

Journal publications

- [J1] **Fast Path Planning Through Large Collections of Safe Boxes** 2024
Tobia Marcucci, Parth Nobel, Russ Tedrake, and Stephen Boyd
IEEE Transactions on Robotics (TRO)
- [J2] **Shortest Paths in Graphs of Convex Sets** 2024
Tobia Marcucci, Jack Umenberger, Pablo A. Parrilo, and Russ Tedrake
SIAM Journal on Optimization
- [J3] **Motion Planning around Obstacles with Convex Optimization** 2023
Tobia Marcucci, Mark Petersen, David von Wrangel, and Russ Tedrake
Science Robotics
- [IEEE RAS TC Model Based Optimization for Robotics Best Paper Award](#)
 - [Cover of November 2023 issue](#)
- [J4] **Warm Start of Mixed-Integer Programs for Model Predictive Control of Hybrid Systems** 2020
Tobia Marcucci and Russ Tedrake
IEEE Transactions on Automatic Control (TAC)
- [J5] **A Two-Stage Trajectory Optimization Strategy for Articulated Bodies with Unscheduled Contact Sequences** 2017
Tobia Marcucci, Marco Gabiccini, and Alessio Artoni
IEEE Robotics and Automation Letters (RAL)

Conference publications

- [C1] **Mixed Discrete and Continuous Planning using Shortest Walks in Graphs of Convex Sets** 2026
Savva Morozov, Tobia Marcucci, Bernhard Paus Graesdal, Alexandre Amice, Pablo Parrilo, and Russ Tedrake
World Symposium on the Algorithmic Foundations of Robotics (WAFR)
- [C2] **A Biconvex Method for Minimum-Time Motion Planning Through Sequences of Convex Sets** 2025
Tobia Marcucci, Mathew Halm, William Yang, Dongchan Lee, and Andrew Marchese
Robotics: Science and Systems (RSS)
- [C3] **A New Semidefinite Relaxation for Linear and Piecewise-Affine Optimal Control with Time Scaling** 2025
Lujie Yang, Tobia Marcucci, Pablo Parrilo, and Russ Tedrake
IEEE International Conference on Robotics and Automation (ICRA)
- [C4] **On the Sample Complexity of Imitation Learning for Smoothed Model Predictive Control** 2024
Daniel Pfrommer, Swati Padmanabhan, Kwangjun Ahn, Jack Umenberger, Tobia Marcucci, Zakaria Mhammedi, and Ali Jadbabaie
IEEE Conference on Decision and Control (CDC)
- [C5] **Multi-Query Shortest-Path Problem in Graphs of Convex Sets** 2024
Savva Morozov, Tobia Marcucci, Alexandre Amice, Bernhard Paus Graesdal, Rohan Bosworth, Pablo

Parrilo, and Russ Tedrake

International Workshop on the Algorithmic Foundations of Robotics (WAFR)

- [C6] **Towards Tight Convex Relaxations for Contact-Rich Manipulation** 2024
Bernhard P. Graesdal, Shao Y.C. Chia, Tobia Marcucci, Savva Morozov, Alexandre Amice, Pablo Parrilo, and Russ Tedrake
Robotics: Science and Systems (RSS)
- [C7] **Approximating Robot Configuration Spaces with few Convex Sets using Clique Covers of Visibility Graphs** 2024
Peter Werner, Alexandre Amice, Tobia Marcucci, Daniela Rus, and Russ Tedrake
IEEE International Conference on Robotics and Automation (ICRA)
- [C8] **Model-Based Control with Sparse Neural Dynamics** 2023
Ziang Liu, Jeff He, Genggeng Zhou, Tobia Marcucci, Li Fei-Fei, Jiajun Wu, and Yunzhu Li
Conference on Neural Information Processing Systems (NeurIPS)
- [C9] **Mixed-Integer Formulations for Optimal Control of Piecewise-Affine Systems** 2019
Tobia Marcucci and Russ Tedrake
ACM International Conference on Hybrid Systems: Computation and Control (HSCC)
- [C10] **Approximate Hybrid Model Predictive Control for Multi-Contact Push Recovery in Complex Environments** 2017
Tobia Marcucci, Robin Deits, Marco Gabiccini, Antonio Bicchi, and Russ Tedrake
IEEE International Conference on Humanoid Robots (Humanoids)
- [C11] **Parametric Trajectory Libraries for Online Motion Planning with Application to Soft Robots** 2017
Tobia Marcucci, Manolo Garabini, Gian Maria Gasparri, Alessio Artoni, Marco Gabiccini, and Antonio Bicchi
International Symposium on Robotic Research (ISRR)
- [C12] **Towards Minimum-Information Adaptive Controllers for Robot Manipulators** 2017
Tobia Marcucci, Cosimo Della Santina, Marco Gabiccini, and Antonio Bicchi
IEEE American Control Conference (ACC)
- [C13] **Approximate Explicit Model Predictive Control for Push Recovery Using Mixed-Integer Convex Optimization** 2017
Robin Deits, Tobia Marcucci, Lucas Manuelli, Twan Koolen, and Russ Tedrake
Dynamic Walking

Preprints

- [P1] **An Efficient Algorithm for Minimum-Pressure Growth Planning of Vine Robots** 2026
Andres C. Torres, Tobia Marcucci, and Elliot W. Hawkes
Under review in IEEE International Conference on Robotics and Automation (ICRA)
arXiv:2609.18070
- [P2] **Biconvex Optimization for Smooth Minimum-Time Trajectories around Convex Obstacles** 2026
Peter Werner, Tobia Marcucci, and Daniela Rus
Under review in IEEE Transactions on Robotics (TRO)
arXiv:2608.02834

- [P3] **A Unified and Scalable Method for Optimization over Graphs of Convex Sets** 2025
Tobia Marcucci
Under review in Mathematical Programming Computation
arXiv:2510.20184

Teaching experience

Main instructor:

- *Optimization and Learning for Robotics and Control* Spring 2026
Graduate course at the University of California, Santa Barbara (ECE 594T)
- *Introduction to Robotics: Planning and Kinematics* Spring 2025, Fall 2025
Undergraduate course at the University of California, Santa Barbara (ECE/ME 179P)

Teaching assistant:

- *Underactuated Robotics* Spring 2020
Graduate course taught by Russ Tedrake at MIT
 - Gave two lectures (available on the [class YouTube channel](#))
 - Developed the exercises in the [class lecture notes](#)
- *Automatic Controls and Robot Mechanics* Fall 2015
Graduate course taught by Antonio Bicchi and Marco Gabiccini at the University of Pisa
 - Gave multiple lectures

Conference and workshop organization

- **World Symposium on the Algorithmic Foundations of Robotics** 2026
Co-chair with Stephane Caron, Steven LaValle, Basak Sakcak, Oren Salzman
- **ScaleOPT: GPU-accelerated and Scalable Optimization** 2025
Workshop at Conference on Neural Information Processing Systems (NeurIPS)
Organizer with Parth Nobel, Fangzhao Zhang, Maximillian Schaller, Tetiana Parshakova, Stephen Boyd
- **Decision and Control Blending Combinatorial and Continuous Optimization** 2023
Workshop at SIAM Conference on Optimization
Main organizer with Jack Umenberger
- **Optimal planning and control fusing offline and online algorithms** 2019
Workshop at IEEE International Conference on Robotics and Automation
Main organizer with Manolo Garabini
- **“Robotics I” session** 2017
IEEE American Control Conference
Co-chair

Invited talks

Motion Planning around Obstacles with Convex Optimization:

- Stanford University (Interactive Perception and Robot Learning Laboratory) 7/2023

- University of California, Berkeley (EECS Seminar) 5/2023
- Stanford University (SystemX Robotics Spotlights) 2/2023
- Cornell University (Verifiable Robotics Group) 10/2022
- Istituto Italiano di Tecnologia (iCub Research Lines) [[recording](#)] 9/2022

Graphs of Convex Sets:

- IEEE International Conference on Robotics and Automation ([Frontiers of Optimization for Robotics Workshop](#)) [[recording](#)] 6/2026
- University of California, Santa Barbara (Applied Math Seminar) 05/2026
- CVXPY Workshop [[recording](#)] 02/2026
- Autonomy Talks [[recording](#)] 11/2025
- University of California, Santa Barbara (ECE Seminar) 11/2025
- International Conference on Continuous Optimization (ICCOPT) (session on “Optimization for robotics”) 7/2025
- INFORMS Annual Meeting (session on “Global optimization”) 10/2023
- SIAM Conference on Optimization (session on “Decision and control blending combinatorial and continuous optimization”) 6/2023
- Stanford University (Linear Algebra and Optimization Seminars) 1/2023
- Joint Mathematics Meetings (SIAM mini-symposium in combinatorial optimization) 1/2023
- International Conference on Optimization and Decision Science (session on “Path and routing problems in industry”) 8/2022
- Université Catholique de Louvain (Cyber-Physical Systems Laboratory) 5/2022
- IMT School for Advanced Studies Lucca 12/2021
- Stanford University (Autonomous Systems Laboratory) 11/2021
- University of California Berkeley (MPC Laboratory) 11/2021
- California Institute of Technology (AMBER Laboratory) 11/2021
- Massachusetts Institute of Technology (Embodied Intelligence Submissions Seminars) 9/2021

Others:

- *Control through Contacts via Approximate Explicit Model Predictive Control* 5/2019
IEEE International Conference on Robotics and Automation
[Workshop on optimal planning and control fusing offline and online algorithms](#)

Invited posters

Graphs of Convex Sets:

- Brown University ([ICERM workshop on Linear and Non-Linear Mixed Integer Optimization](#)) 2/2023
- Cornell University ([ORIE Young Researchers Workshop](#)) 10/2022

Awards

- [MIT EECS George M. Sprowls Ph.D. Thesis Award in Artificial Intelligence and Decision Making](#) 2025
- [IEEE RAS TC Model Based Optimization for Robotics Best Paper Award](#) 2023
- [SIAM Student Travel Award](#) 2023
- Grass Instruments Company Fellow 9/2018 to 5/2019

Reviewer

- International journals and conferences, including: Automatica, IEEE Transactions on Automatic Control (TAC), IEEE Transactions on Robotics (TRO), International Journal of Robotics Research (IJRR), Journal of Robust and Nonlinear Control, Journal of Optimization Theory and Applications (JOTA), and Science Robotics.